

mentalimpairment-code.r

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```
setwd("~/Dropbox/Classes/536/CaseStudies/MentalImpairment")

data <- read.table("mentalimpairment-data.txt",header=TRUE)
attach(data)
```

#first do not take into consideration the order

```
library(nnet)
(MO = multinom(Impairment ~ 1))
```

```
## # weights:  8 (3 variable)
## initial  value 55.451774
## final   value 54.521026
## converged

## Call:
## multinom(formula = Impairment ~ 1)
##
## Coefficients:
##             (Intercept)
## Mild             0.2876842
## Moderate        -0.2513121
## Well             0.2876842
##
## Residual Deviance: 109.0421
## AIC: 115.0421
```

```
summary(MO)
```

```
## Call:
## multinom(formula = Impairment ~ 1)
##
## Coefficients:
##             (Intercept)
## Mild             0.2876842
## Moderate        -0.2513121
## Well             0.2876842
##
## Std. Errors:
##             (Intercept)
## Mild             0.4409587
## Moderate         0.5039527
## Well             0.4409587
##
## Residual Deviance: 109.0421
## AIC: 115.0421
```

```
BIC(MO)
```

```
## [1] 120.1087
```

```
(Ms = multinom(Impairment ~ SES))
```

```
## # weights: 12 (6 variable)
## initial value 55.451774
## final value 52.642355
## converged

## Call:
## multinom(formula = Impairment ~ SES)
##
## Coefficients:
##          (Intercept)          SES
## Mild      -2.231483e-01  0.9163037
## Moderate  -8.032857e-06 -0.6931261
## Well      -2.231480e-01  0.9163039
##
## Residual Deviance: 105.2847
## AIC: 117.2847
```

```
summary(Ms)
```

```
## Call:
## multinom(formula = Impairment ~ SES)
##
## Coefficients:
##          (Intercept)          SES
## Mild      -2.231483e-01  0.9163037
## Moderate  -8.032857e-06 -0.6931261
## Well      -2.231480e-01  0.9163039
##
## Std. Errors:
##          (Intercept)          SES
## Mild      0.6708198 0.9082957
## Moderate  0.6324555 1.0723800
## Well      0.6708198 0.9082956
##
## Residual Deviance: 105.2847
## AIC: 117.2847
```

```
BIC(Ms)
```

```
## [1] 127.418
```

```
(Me = multinom(Impairment ~ Events))
```

```
## # weights: 12 (6 variable)
## initial value 55.451774
## iter 10 value 50.908223
## final value 50.908198
## converged

## Call:
## multinom(formula = Impairment ~ Events)
##
## Coefficients:
##          (Intercept)      Events
## Mild      1.673461 -0.2673717
```

```
## Moderate      1.202394 -0.2835592
## Well          2.453627 -0.4847294
##
## Residual Deviance: 101.8164
## AIC: 113.8164
```

```
summary(Me)
```

```
## Call:
## multinom(formula = Impairment ~ Events)
##
## Coefficients:
##      (Intercept)      Events
## Mild          1.673461 -0.2673717
## Moderate      1.202394 -0.2835592
## Well          2.453627 -0.4847294
##
## Std. Errors:
##      (Intercept)      Events
## Mild          1.046235 0.1760685
## Moderate      1.143202 0.2012544
## Well          1.049018 0.2014680
##
## Residual Deviance: 101.8164
## AIC: 113.8164
```

```
BIC(Me)
```

```
## [1] 123.9497
```

```
(M = multinom(Impairment ~ SES + Events))
```

```
## # weights: 16 (9 variable)
## initial value 55.451774
## iter 10 value 48.350248
## final value 48.349131
## converged

## Call:
## multinom(formula = Impairment ~ SES + Events)
##
## Coefficients:
##      (Intercept)      SES      Events
## Mild          1.247474  1.3575833 -0.3316398
## Moderate      1.240500 -0.3374756 -0.2671694
## Well          1.902708  1.6062981 -0.5601850
##
## Residual Deviance: 96.69826
## AIC: 114.6983
```

```
summary(M)
```

```
## Call:
## multinom(formula = Impairment ~ SES + Events)
##
## Coefficients:
##      (Intercept)      SES      Events
```

```
## Mild      1.247474  1.3575833 -0.3316398
## Moderate  1.240500 -0.3374756 -0.2671694
## Well      1.902708  1.6062981 -0.5601850
##
## Std. Errors:
##      (Intercept)      SES      Events
## Mild      1.104013  1.014488  0.1925961
## Moderate  1.154363  1.135453  0.2073255
## Well      1.110767  1.080474  0.2190439
##
## Residual Deviance: 96.69826
## AIC: 114.6983
```

```
BIC(M)
```

```
## [1] 129.8982
```

```
1-pchisq(M0$deviance-Me$deviance,length(coef(Me))-length(coef(M0)))
```

```
## [1] 0.06504276
```

```
BIC(M0)
```

```
## [1] 120.1087
```

```
BIC(Ms)
```

```
## [1] 127.418
```

```
BIC(Me)
```

```
## [1] 123.9497
```

```
BIC(M)
```

```
## [1] 129.8982
```

```
AIC(M0)
```

```
## [1] 115.0421
```

```
AIC(Ms)
```

```
## [1] 117.2847
```

```
AIC(Me)
```

```
## [1] 113.8164
```

```
AIC(M)
```

```
## [1] 114.6983
```

```
#now take the order into account
```

```
library(MASS)
```

```
Impairment = factor(Impairment,levels=c("Well","Mild","Moderate","Impaired"))
```

```
(orderM0 = polr(Impairment ~ 1))
```

```
## Call:
```

```
## polr(formula = Impairment ~ 1)
```

```
##
```

```
## No coefficients
##
## Intercepts:
##      Well|Mild      Mild|Moderate Moderate|Impaired
##      -0.8472977      0.4054527      1.2367503
##
## Residual Deviance: 109.0421
## AIC: 115.0421
summary(orderM0)
```

```
##
## Re-fitting to get Hessian
## Call:
## polr(formula = Impairment ~ 1)
##
## No coefficients
##
## Intercepts:
##      Value Std. Error t value
## Well|Mild -0.8473  0.3450  -2.4557
## Mild|Moderate 0.4055  0.3227   1.2562
## Moderate|Impaired 1.2368  0.3786   3.2663
##
## Residual Deviance: 109.0421
## AIC: 115.0421
```

```
BIC(orderM0)
```

```
## [1] 120.1087
```

```
(orderMs = polr(Impairment ~ SES))
```

```
## Call:
## polr(formula = Impairment ~ SES)
##
## Coefficients:
##      SES
## -0.8553378
##
## Intercepts:
##      Well|Mild      Mild|Moderate Moderate|Impaired
##      -1.36376017      -0.04172851      0.83061097
##
## Residual Deviance: 106.8744
## AIC: 114.8744
```

```
summary(orderMs)
```

```
##
## Re-fitting to get Hessian
## Call:
## polr(formula = Impairment ~ SES)
##
## Coefficients:
##      Value Std. Error t value
```

```
## SES -0.8553      0.5865  -1.458
##
## Intercepts:
##           Value Std. Error t value
## Well|Mild    -1.3638  0.5038   -2.7070
## Mild|Moderate -0.0417  0.4448   -0.0938
## Moderate|Impaired 0.8306  0.4659    1.7827
##
## Residual Deviance: 106.8744
## AIC: 114.8744
BIC(orderMs)

## [1] 121.6299
(orderMe = polr(Impairment ~ Events))

## Call:
## polr(formula = Impairment ~ Events)
##
## Coefficients:
## Events
## 0.28793
##
## Intercepts:
##           Well|Mild      Mild|Moderate Moderate|Impaired
##           0.2614334          1.6562752          2.5876265
##
## Residual Deviance: 102.5271
## AIC: 110.5271
summary(orderMe)

##
## Re-fitting to get Hessian
## Call:
## polr(formula = Impairment ~ Events)
##
## Coefficients:
##           Value Std. Error t value
## Events 0.2879      0.1175    2.451
##
## Intercepts:
##           Value Std. Error t value
## Well|Mild      0.2614 0.5639    0.4636
## Mild|Moderate   1.6563 0.6171    2.6838
## Moderate|Impaired 2.5876 0.6938    3.7296
##
## Residual Deviance: 102.5271
## AIC: 110.5271
BIC(orderMe)

## [1] 117.2826
(orderM = polr(Impairment ~ SES + Events))
```

```
## Call:
## polr(formula = Impairment ~ SES + Events)
##
## Coefficients:
##      SES      Events
## -1.1112310  0.3188613
##
## Intercepts:
##      Well|Mild      Mild|Moderate Moderate|Impaired
##      -0.2819031      1.2127926      2.2093721
##
## Residual Deviance: 99.0979
## AIC: 109.0979
```

```
summary(orderM)
```

```
##
## Re-fitting to get Hessian
## Call:
## polr(formula = Impairment ~ SES + Events)
##
## Coefficients:
##      Value Std. Error t value
## SES      -1.1112      0.6109  -1.819
## Events  0.3189      0.1210   2.635
##
## Intercepts:
##      Value Std. Error t value
## Well|Mild      -0.2819  0.6423  -0.4389
## Mild|Moderate      1.2128  0.6607   1.8357
## Moderate|Impaired  2.2094  0.7210   3.0644
##
## Residual Deviance: 99.0979
## AIC: 109.0979
```

```
BIC(orderM)
```

```
## [1] 117.5423
```

```
#get the slopes
```

```
coef(orderM)
```

```
##      SES      Events
## -1.1112310  0.3188613
```

```
#get the intercepts
```

```
orderM$zeta
```

```
##      Well|Mild      Mild|Moderate Moderate|Impaired
##      -0.2819031      1.2127926      2.2093721
```

```
#fitted probabilities
```

```
orderM$fitted
```

```
##      Well      Mild      Moderate      Impaired
## 1  0.62491585 0.2564222 0.07131474 0.04734725
## 2  0.11502358 0.2518325 0.24398557 0.38915837
## 3  0.39028430 0.3502175 0.14495563 0.11454256
```

```

## 4 0.46822886 0.3287373 0.11707594 0.08595788
## 5 0.28503416 0.3548980 0.18808642 0.17198137
## 6 0.69621280 0.2146344 0.05428168 0.03487110
## 7 0.35416883 0.3555319 0.15911298 0.13118624
## 8 0.46822886 0.3287373 0.11707594 0.08595788
## 9 0.46822886 0.3287373 0.11707594 0.08595788
## 10 0.19738781 0.3255946 0.22513070 0.25188691
## 11 0.35416883 0.3555319 0.15911298 0.13118624
## 12 0.28503416 0.3548980 0.18808642 0.17198137
## 13 0.31756633 0.3571768 0.17419493 0.15106190
## 14 0.10019399 0.2315350 0.24178533 0.42648565
## 15 0.46822886 0.3287373 0.11707594 0.08595788
## 16 0.35416883 0.3555319 0.15911298 0.13118624
## 17 0.15167000 0.2918553 0.23993344 0.31654127
## 18 0.54775531 0.2959818 0.09227179 0.06399113
## 19 0.13282485 0.2729369 0.24333367 0.35090454
## 20 0.31756633 0.3571768 0.17419493 0.15106190
## 21 0.11502358 0.2518325 0.24398557 0.38915837
## 22 0.22469945 0.3390046 0.21407808 0.22221786
## 23 0.46822886 0.3287373 0.11707594 0.08595788
## 24 0.62491585 0.2564222 0.07131474 0.04734725
## 25 0.42998727 0.3408054 0.13029529 0.09891202
## 26 0.39028430 0.3502175 0.14495563 0.11454256
## 27 0.22469945 0.3390046 0.21407808 0.22221786
## 28 0.04102612 0.1191445 0.18048372 0.65934567
## 29 0.25278003 0.3485130 0.20206806 0.19663890
## 30 0.17402747 0.3103145 0.23352933 0.28212871
## 31 0.22469945 0.3390046 0.21407808 0.22221786
## 32 0.15167000 0.2918553 0.23993344 0.31654127
## 33 0.54775531 0.2959818 0.09227179 0.06399113
## 34 0.19738781 0.3255946 0.22513070 0.25188691
## 35 0.13282485 0.2729369 0.24333367 0.35090454
## 36 0.17402747 0.3103145 0.23352933 0.28212871
## 37 0.17402747 0.3103145 0.23352933 0.28212871
## 38 0.15167000 0.2918553 0.23993344 0.31654127
## 39 0.05557758 0.1522454 0.20761767 0.58455931
## 40 0.04102612 0.1191445 0.18048372 0.65934567

```